

Workshop SageMaker Unified Studio Setup Guide

Deployment preparation and validation guide for the workshop SMUS workshop environment

Purpose and Scope

This guide summarizes the actual setup path used for the SageMaker Unified Studio workshop: IAM Identity Center access, CloudFormation baseline deployment, domain/profile setup, project creation, S3/Glue/Lake Formation access, and validation. Items inside SMUS are included only where required for deployment validation.

Item	Value
Account / Region	<insert target account>
Domain	workshop-smus-workshop
Domain URL	<insert sagemaker domain url>
Project / Profile	workshop-workshop-poc
Project storage bucket	s3://amazon-sagemaker-<target-account-id>-ap-southeast-1-workshop-smus
Data source bucket	s3://workshop-workshop-bucket/

Preparation

- Confirm target account, Region, domain name, deployment tags, and private subnet CIDRs.
- Confirm IAM Identity Center users/groups for domain access and project membership.
- Update the workshop-Sandbox-GlueAccess permission set with required analytics services and CloudShell access (cloudshell:*); re-provision the account assignment.
- Review services-access.yaml and confirm the project storage bucket is separate from the existing workshop data source bucket.

CloudFormation Deployment

Area	Deployed / Configured
Networking	Dedicated VPC, 3 private subnets, private route table, route associations, endpoint security group.
Endpoints	S3 Gateway plus STS, DataZone, Glue, Athena, SageMaker API, KMS, and Secrets Manager interface endpoints.
Storage	SMUS project bucket with versioning, encryption, and public access block.
IAM roles	DomainExecution, DomainService, AdminIAMExecution, Provisioning, QueryExecution, S3Tables LF role, and ManageAccess roles.
SMUS/DataZone	Domain, SSO user profiles, Jemima root domain owner, and Tooling/DataLake/LakehouseCatalog blueprint configs.
Data source access	Active SMUS project role granted list/read access to approved workshop-workshop-bucket prefixes.

1. Deploy or update the stack in CloudFormation using services-access.yaml.
2. Review the change set, especially IAM, S3, networking, DataZone, and Lake Formation-related changes.
3. Wait for CREATE_COMPLETE or UPDATE_COMPLETE and capture stack outputs for handoff.

SMUS Project and Data Setup

Step	Configuration / Expected Result
Domain access	Open the SageMaker application matching dzd-beaiy3asjwltcx from AWS Access Portal.
Project profile	Use the workshop project profile; deploy in the target account and ap-southeast-1 region; set S3 storage to the SMUS project bucket.
Project	Create workshop-workshop-poc and add required SSO users/group as project members.
S3 connection	Add bronze/silver/gold/quarantine prefixes from workshop-workshop-bucket as needed.
Glue source	Create AWS Glue (Lakehouse) source using project.default_lakehouse, AwsDataCatalog, DB workshop_bronze, include sales_raw*.

Validation	Run the data source and confirm sales_raw is created/searchable under workshop_bronze.
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Lake Formation Grants Applied

- Added AmazonSageMakerDomainExecution as Lake Formation data lake administrator.
- Added AWSReservedSSO_workshop-Sandbox-GlueAccess_1437bb9f121dab96 as Lake Formation data lake administrator for CloudShell troubleshooting.
- Granted DESCRIBE on workshop_bronze to datazone_usr_role_bx6wr6wcds35wh_4mmd2khfn7cmn5.
- Granted DESCRIBE and SELECT on workshop_bronze.sales_raw to the same project user role.

Issues Encountered and Resolution

Issue	Resolution
Wrong SMUS page / SsoLogin denied	Use the correct AWS Access Portal application and confirm SSO user assignment.
Blueprint authorization error	Authorize the correct domain unit/project profile for Tooling/Lakehouse blueprints.
s3:PutObject to data bucket	Use the SMUS project bucket for project files, not workshop-workshop-bucket.
Missing role config	Recreate/update project profile with valid Tooling/user role configuration.
No project access	Add users or SSO group under project Members.
S3 ListBucket denied	Grant the active project user role ListBucket/GetObject to approved source prefixes.
Lake Formation grant denied	Add the caller SSO role as data lake admin, then rerun grants.
Glue source Added 0	Use wildcard once for first import if needed, then narrow back to sales_raw*.
Failed duplicate DB entries	Use existing Glue databases through data source onboarding; validate actual resources in Glue/Lake Formation.

Final Validation Checklist

- CloudFormation stack is complete and outputs are recorded.
- Users can open the correct SMUS domain and project.
- S3 bronze prefix lists successfully in SMUS.
- Glue database workshop_bronze and table sales_raw exist.
- Data source workshop-bronze-sales-raw-glue completes; sales_raw asset is searchable.

References

- Create SMUS domain manually:
<https://docs.aws.amazon.com/sagemaker-unified-studio/latest/adminguide/create-domain-sagemaker-unified-studio-manual.html>
- AmazonSageMakerDomainExecution:
<https://docs.aws.amazon.com/sagemaker-unified-studio/latest/adminguide/AmazonSageMakerDomainExecution.html>
- AmazonSageMakerDomainService:
<https://docs.aws.amazon.com/sagemaker-unified-studio/latest/adminguide/AmazonSageMakerDomainService.html>